

# Commercialization of Graphene Applications Through Malaysia's National Graphene Action Plan 2020 (NGAP2020)



**Graphene  
& 2D Materials** 

Date : 18th – 19<sup>th</sup> November 2015  
Venue : Santa Clara Convention Center, CA





## Statement by Prime Minister

*"Nanotechnology development would be given and be made of the resources of the country's new economic model. Thus, it is important for Malaysia to not be left behind in the field of nanotechnology and we have decided to give importance...."*

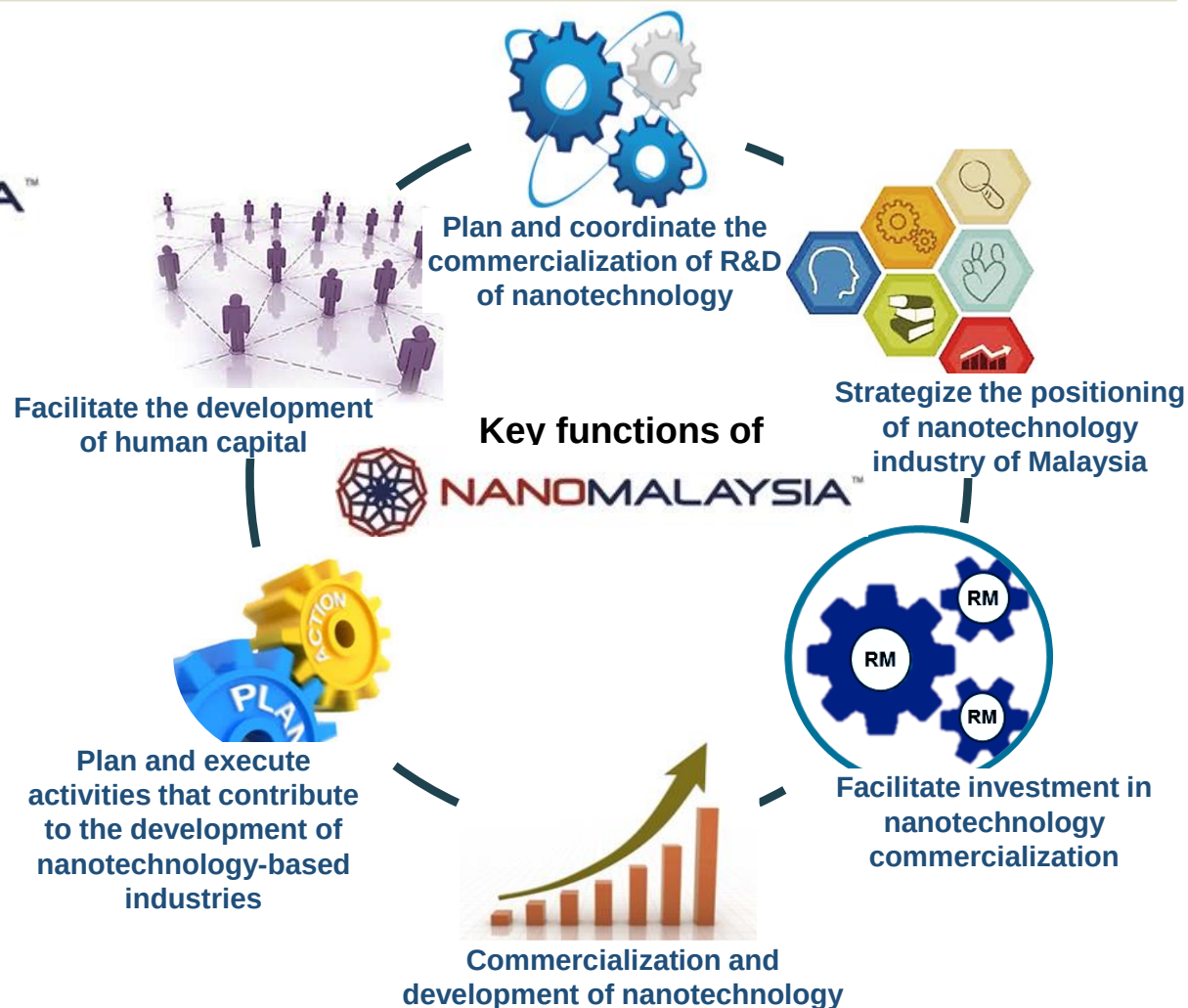
**YAB Dato' Sri Mohd Najib Tun Razak**

29 Oct 2009



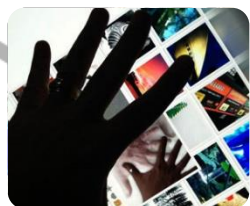
**NanoMalaysia Berhad (NanoMalaysia)** was mooted during **National Innovation Council (NIC) meeting**, chaired by the Right Honorable Prime Minister, Y.A.B. Dato' Seri Najib Tun Razak on **14 February 2011**.

NanoMalaysia was then incorporated as Company Limited by Guarantee (CLG) in August 2011 under Section 24, Company Act 1965 to take the lead in **commercialisation of nanotechnology products**



## Two (2) Commercialization Programs in Nanotechnology

### 1 iNanovation



Technology and  
Business Due  
Diligence



Nanotechnology  
landscaping and  
Business  
opportunities



Strategy planning &  
consulting in  
commercialization of  
Nanotechnology R&D



Facilitation of  
investment in  
Nanotechnology



Development of  
human capitals in  
Nanotechnology



Food &  
Agriculture



Electronics,  
Devices &  
Systems



Energy &  
Environment



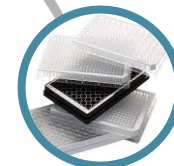
Wellness,  
Medical &  
Healthcare

### 2

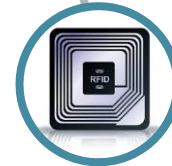
### National Graphene Action Plan 2020 (NGAP2020)



Nanofluid



Plastic  
additives



Condu-  
ctive ink



Li-ion  
battery/  
ultra-  
capacitors



Rubber  
additives



**NANOVerify**  
Programme

Nanotechnology Product and Process  
Verification and Certification supporting  
iNanovation and NGAP 2020



# Highlights of the National Graphene Action Plan 2020 (NGAP2020)

## National Graphene Action Plan 2020

- Bridging the gap: Industry and Research Academia
- Encourage industry to move up the value chain by introducing high technology products



## Highlights

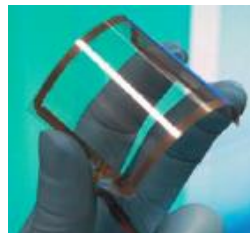
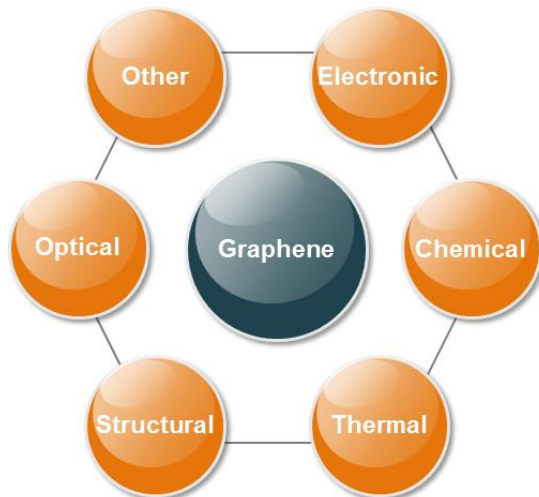
- 1 “High value-add manufacturing enabled by Graphene with IP in selected application areas”
- 2 Enhancing downstream applications relevant to Malaysia; and eventually enabling a local Graphene eco-system to accelerate downstream adoption
- 3 Prioritization of five application areas;
  - ✓ Lithium-ion battery anodes / ultracapacitors
  - ✓ Conductive inks
  - ✓ Rubber additives
  - ✓ Plastics additives
  - ✓ Nanofluids
- 4 By 2020, the National Graphene Action Plan 2020 has the potential to add more than RM 20 Bn in GNI impact and help create 9,000 Malaysian jobs



# Graphene, widely known as 'wonder material' has superior physical properties and could bring immense potential to intensify MY economic growth via the manufacturing sector

Graphene is one of the most important innovations in material sciences in the last decade

- First isolated in 2004, Graphene is a one atomic layer thick honeycomb sheet of carbon atoms.
- Several superior properties have already been proven, and new properties and applications are continually being discovered.
- Leading countries have already invested in innovation, further building up global momentum behind R&D.



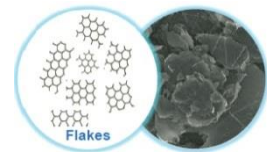
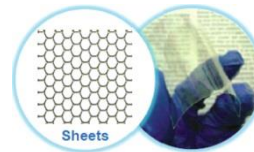
Sheet Graphene



Flake Graphene

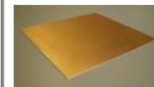
Forms and methods of production varies for different applications

- Graphene is commonly available in two forms; i) sheets, and ii) flakes.



- **Chemical Vapor Deposition (CVD)** and **Chemical Exfoliation (CE)** are the common methods of Graphene production.

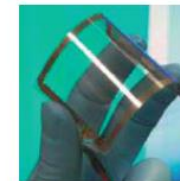
Chemical Vapor Deposition



Sample substrate



CVD Apparatus



Sheet Graphene

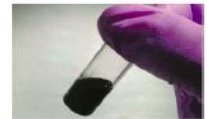
Chemical Exfoliation



Block of Graphite



Ultrasonic Bath



Flake Graphene

# Background of the National Graphene Action Plan 2020 (NGAP2020)

## From feasibility study to execution of the Action Plan...

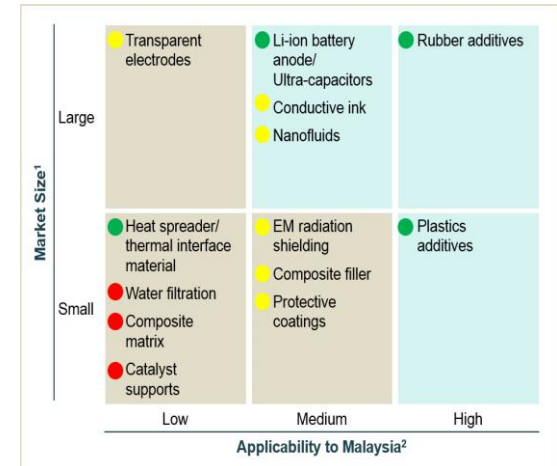
Feasibility study on Graphene downstream applications relevant to Malaysia

1

Based on **technical feasibility, economic viability, time to adoption, relevance and impact** to Malaysia (2020 timeframe), **five (5) application areas** were prioritized

2

- Lithium ion battery
- Rubber additives
- Plastic additives
- Conductive Ink
- Nanofluids



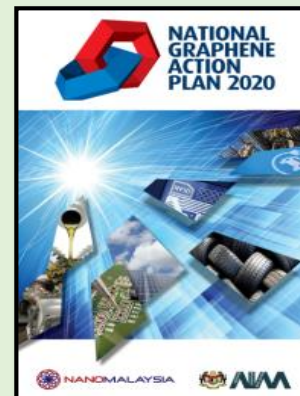
Apr 2014



NANOMALAYSIA™

Jul 2014

Launch of  
NGAP 2020

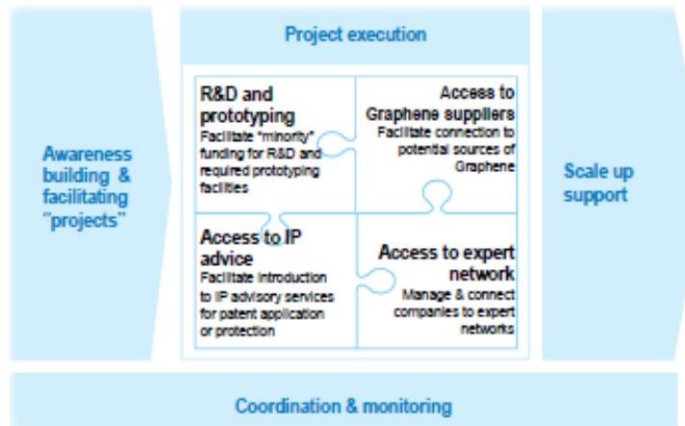


3

▪ NanoMalaysia appointed as the Lead agency to execute the NGAP2020

Role includes:

- Awareness building
- Project execution
- Scale support
- Coordination & Monitoring



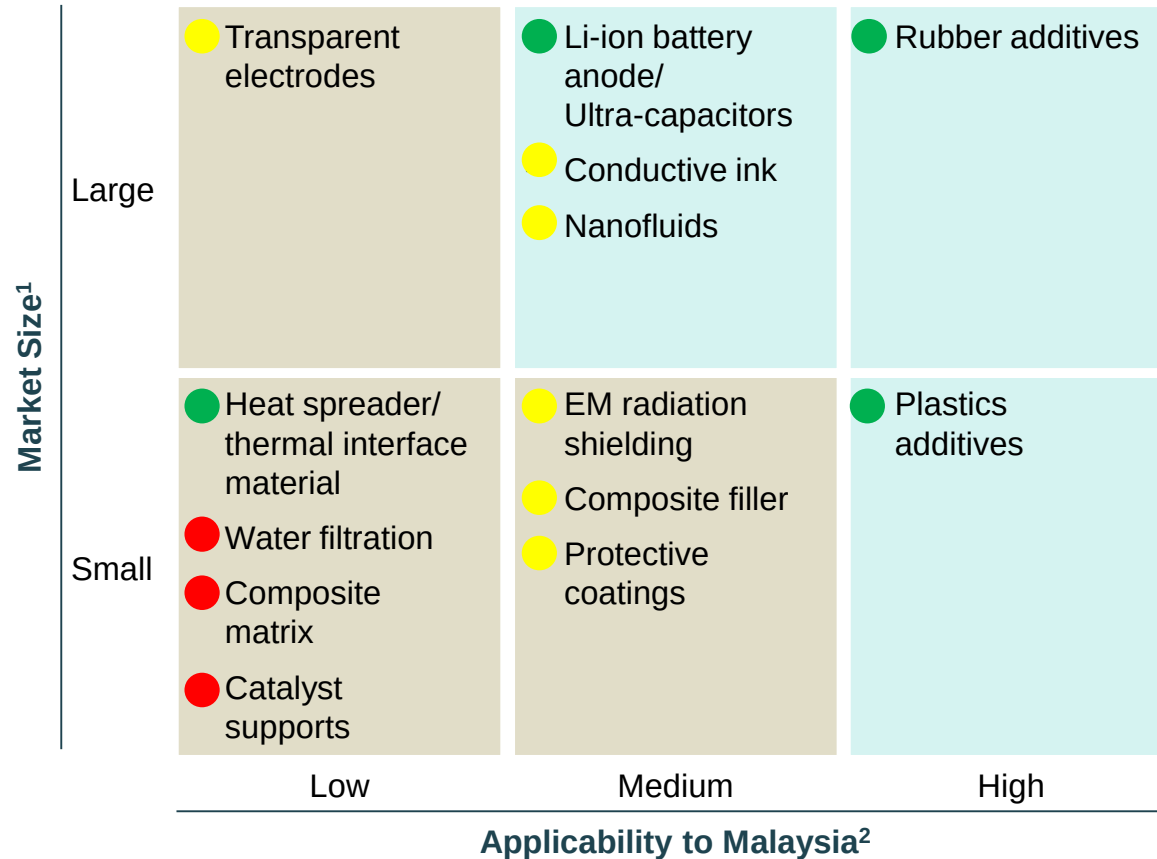
# Through systematic analysis we narrowed down to 5 technologically and economically feasible applications ...

## Market size and applicability to Malaysia

Likelihood of successful Graphene commercialization

● High ● Medium ● Low

■ For further consideration



### The five (5) applications areas for further consideration are:

- Li-ion battery anode / ultra-capacitors
- Conductive inks
- Rubber additives
- Plastic additives
- Nano fluids including drilling fluids, lubricants and coolants

1 Market size is high if addressed Graphene market size is ~\$1B or higher

2 Applicability to MY is based on current downstream players, upcoming policies, and focus of neighboring countries

# There is significant value in catalyzing downstream differentiation through adoption of Graphene

■ Addressable market size  
■ Total downstream market size

## Graphene and application market size for 2020<sup>1</sup>

USD Billions



<sup>1</sup> Estimated global sales of the applications which would be Graphene enabled, and estimated global sales of the Graphene integrated into these applications



# Overview of potential downstream applications for graphene.

## Major applications

|                                                                                    |                                                                                                                                 |                                                                                     |                                                                                                                                                  |                                                                                       |                                                                                                                 |                                                                                       |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|    | <b>Electric Vehicles</b> <ul style="list-style-type: none"> <li>▪ Hybrid Evs</li> <li>▪ Plug-in HEVs</li> <li>▪ BEVs</li> </ul> |    | <b>Small IT</b> <ul style="list-style-type: none"> <li>▪ Smartphones</li> <li>▪ Tablets</li> <li>▪ Notebook PCs</li> </ul>                       |    | <b>Energy Storage</b> <ul style="list-style-type: none"> <li>▪ Load shifting</li> <li>▪ Grid support</li> </ul> |    |
|    | <b>Photo-voltaics</b> <ul style="list-style-type: none"> <li>▪ Photovoltaic-cells</li> <li>▪ Bus bars</li> </ul>                |    | <b>Medical</b> <ul style="list-style-type: none"> <li>▪ Electrodes</li> <li>▪ Transdermal drug delivery</li> </ul>                               |    | <b>Others</b> <ul style="list-style-type: none"> <li>▪ RFID</li> <li>▪ PCBs</li> <li>▪ E-packaging</li> </ul>   |    |
|    | <b>Latex</b> <ul style="list-style-type: none"> <li>▪ Gloves</li> <li>▪ Condoms</li> </ul>                                      |    | <b>Tires</b> <ul style="list-style-type: none"> <li>▪ Cars, trucks</li> <li>▪ Off-highway</li> </ul>                                             |    | <b>Rubber goods</b> <ul style="list-style-type: none"> <li>▪ Industrial equipment</li> <li>▪ Auto</li> </ul>    |    |
|   | <b>HPP</b> <ul style="list-style-type: none"> <li>▪ PEEK</li> </ul>                                                             |   | <b>Engineering Plastics</b> <ul style="list-style-type: none"> <li>▪ PC</li> <li>▪ PET</li> </ul>                                                |   | <b>Commodity Plastics</b> <ul style="list-style-type: none"> <li>▪ PVC</li> <li>▪ ABS</li> </ul>                |   |
|  | <b>Drilling fluid</b> <ul style="list-style-type: none"> <li>▪ Oil recovery</li> </ul>                                          |  | <b>Lubricants</b> <ul style="list-style-type: none"> <li>▪ Gear lubricants</li> <li>▪ Bearing lubricant</li> <li>▪ Cylinder lubricant</li> </ul> |  | <b>Others</b> <ul style="list-style-type: none"> <li>▪ Coolant</li> </ul>                                       |  |

# Alignment with NKEAs of Economic Transformation Programme (ETP)



**Palm Oil  
and Rubber**

1



**Food & Agriculture**



Incremental GNI  
by 2020  
**RM29 billion**



Additional Jobs  
by 2020  
**109,335**

**EPP Highlights**

9.3

Increase World Market Share of  
Latex Gloves to 65% by 2020



**Electrical &  
Electronics**

2



**Electronics,  
Devices & Systems**



Incremental GNI  
by 2020  
**RM53 billion**



Additional Jobs  
by 2020  
**157,000**

**EPP Highlights**

20

Enabling industries through  
Nanotechnology



**Oil, Gas  
and Energy**

3



**Energy &  
Environment**



Incremental GNI  
by 2020  
**RM131 billion**



Additional Jobs  
by 2020  
**52,300**

**EPP Highlights**

1

Rejuvenating Existing Fields through  
Enhanced Oil Recovery (EOR)

10

Increase Petrochemical Outputs

7

Increasing solar module producers

18

Enable EV component manufacturing



**Healthcare**

4



**Wellness, Medical  
& Healthcare**



Incremental GNI  
by 2020  
**RM35 billion**



Additional Jobs  
by 2020  
**181,000**

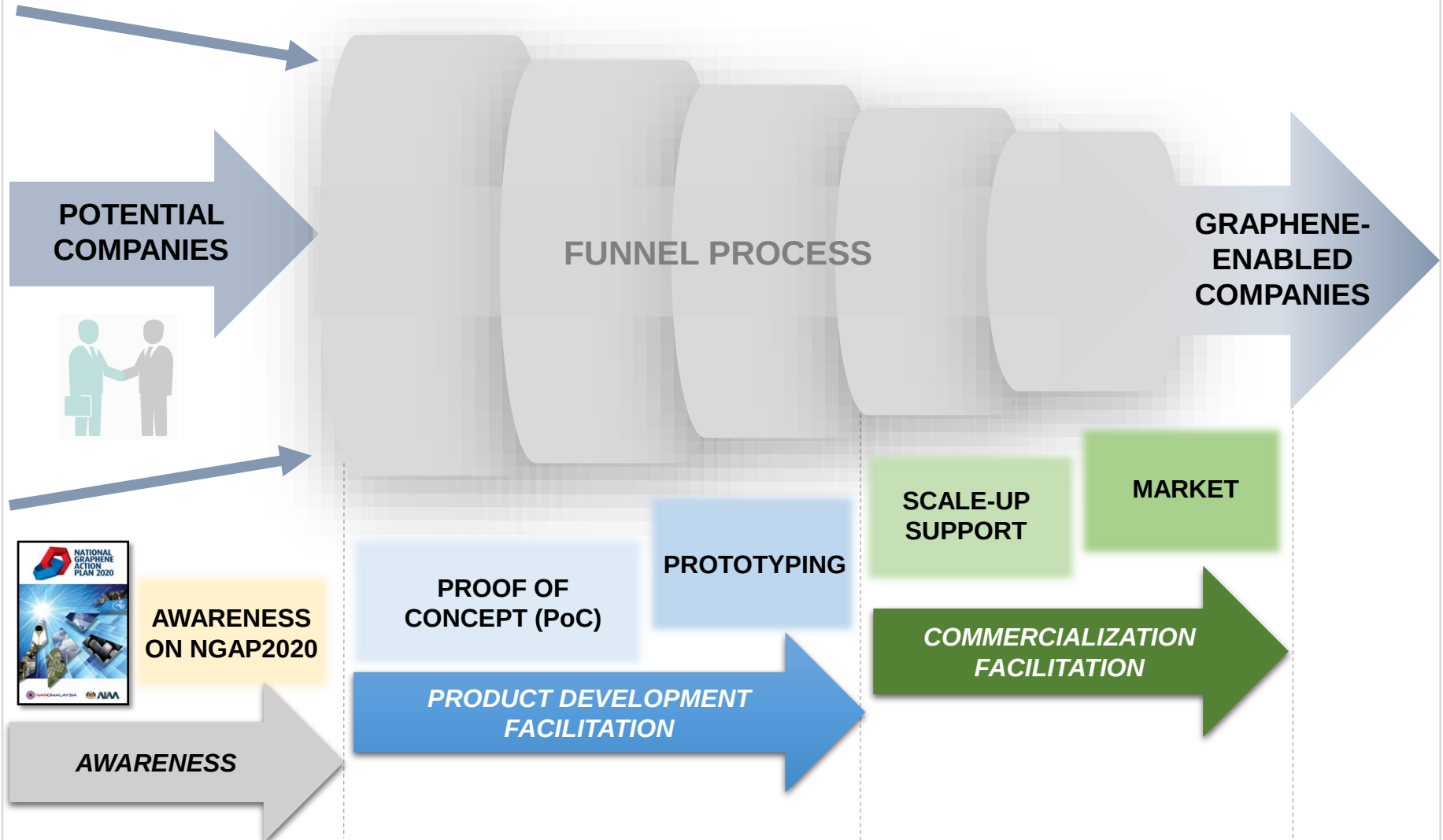
**EPP Highlights**

2

Creating supportive ecosystem to  
grow clinical research

# Delivery Framework for National Graphene Action Plan 2020

*Industry Engagement and Project Facilitation is based on the identified framework, with supporting activities to ensure project delivery*



# Project Facilitation by NGAP2020

## AWARENESS & INITIAL DISCUSSION

- Involve feasibility study on business and technology viability of product when incorporated with graphene.
- Proposal (5-pager) submission by companies on developmental activities related to graphene.
- Handholding on technical / process review to the company include:



- ✓ Technical sourcing
- ✓ Graphene samples (preliminary trials)
- ✓ Formulation and methodology statement
- ✓ Pre-manufacturing development



## PRODUCT DEVELOPMENT

- PoC
- Prototyping

- Provide NGAP's catalytic funding (upon endorsement of NGAP's Steering Committee Members) to support:
  - ✓ Production of samples for PoC and Prototyping
  - ✓ Sample testing
  - ✓ Customer trials



- NGAP's catalytic funding include:
  - ✓ Technical consultation (access to technical experts)
  - ✓ Graphene supply
  - ✓ Equipment rentals for testing
- Access to other MOSTI's R&D funding.
- Roundtable – Graphene Investment Opportunities (R-GIO)
- Access to IP advice



## COMMERCIALIZATION

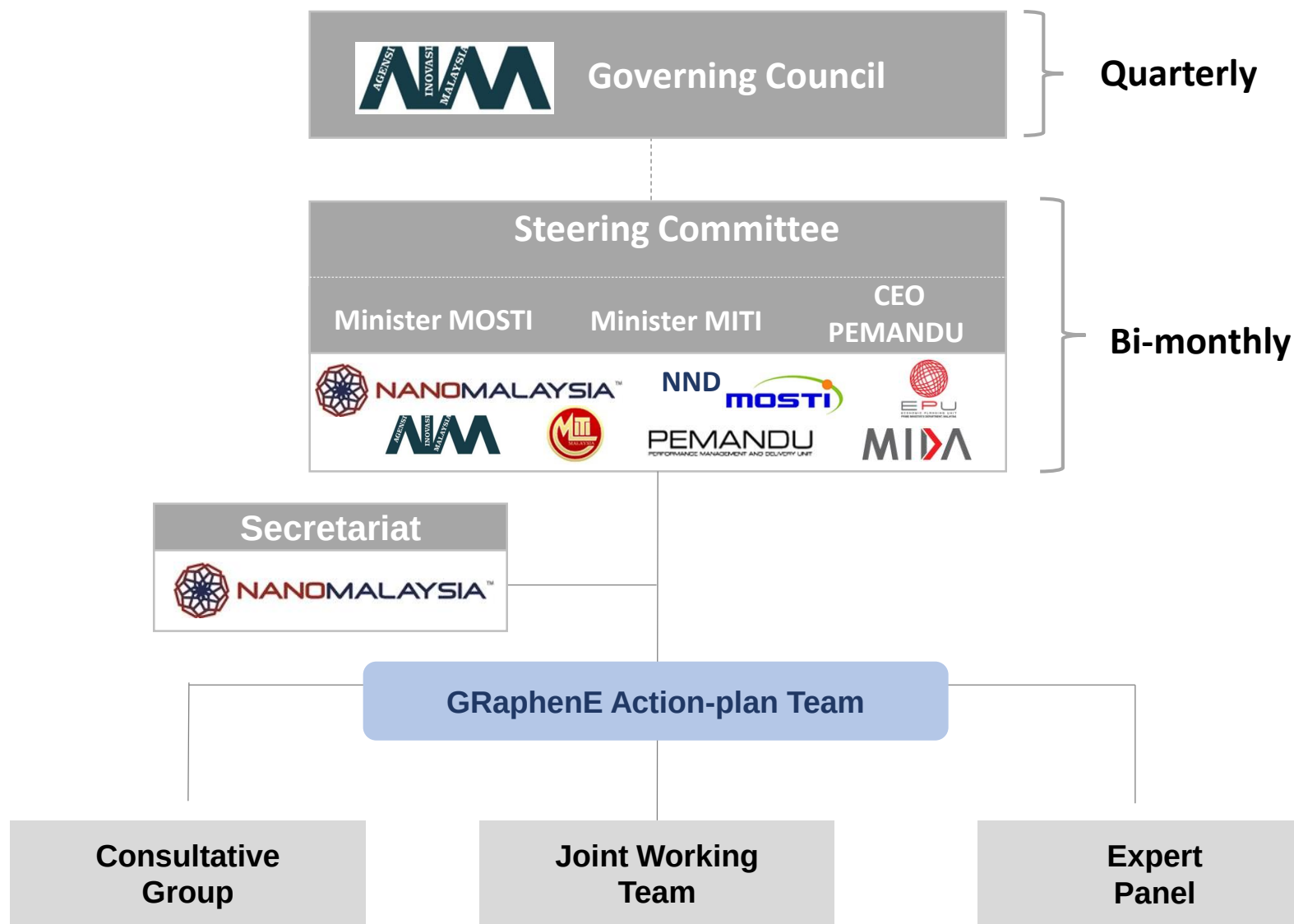
- Scale-up support
- Market

- Provide facilitation on:
  - ✓ Access to IP advice
  - ✓ Matching to Government's assistance (MIDA, PEMANDU, AIM, PEs, VCs, etc.) for funds / incentives support and facilitate to suitable EPP-ready projects.
  - ✓ Roundtable – Graphene Investment Opportunities (R-GIO)





# Governance structure in place to execute the National Graphene Action Plan 2020 (NGAP2020)



# Overview of Product Development Projects under NGAP2020 Program

## National target by 2020

- RM 20 Bn in GNI impact
- 9000 new Malaysian jobs

## Industry target

- Increase market share (local and global)
- New market penetration
- Retain market competitiveness

### Conductive Ink

Potential application in:

- Electronic packaging
- Smart card



### Nanofluid

Potential application in:

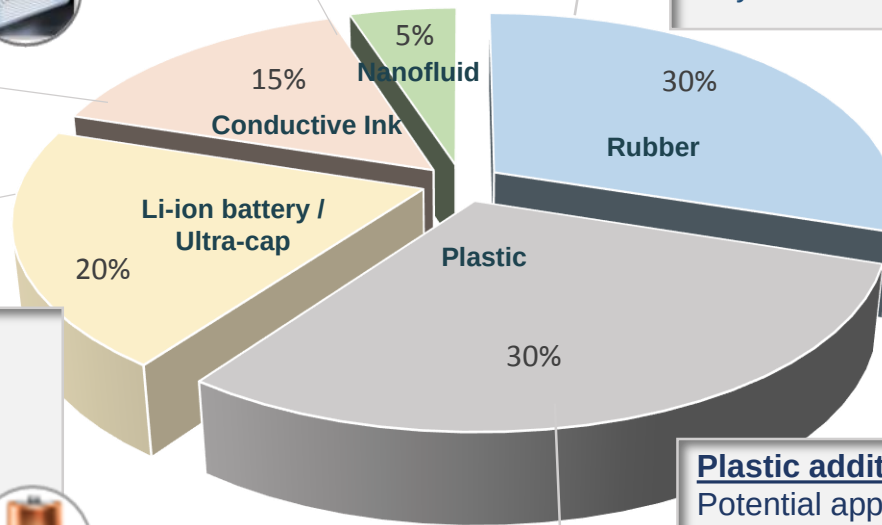
- Lubricants



### Rubber additives

Potential application in:

- Glove
- Condom
- Tyre
- Specialty hose
- Masterbatch



### Li-ion battery /ultracapacitor

Potential application in:

- Electric vehicle
- Wearable devices



### Plastic additives

Potential application in:

- Automotive component
- Composite
- Glove
- Electronics
- Solar Panel



Percentage of Companies



Li-ion battery/  
ultra-capacitors



Conductive  
inks



Plastics  
additives



Rubber  
additives



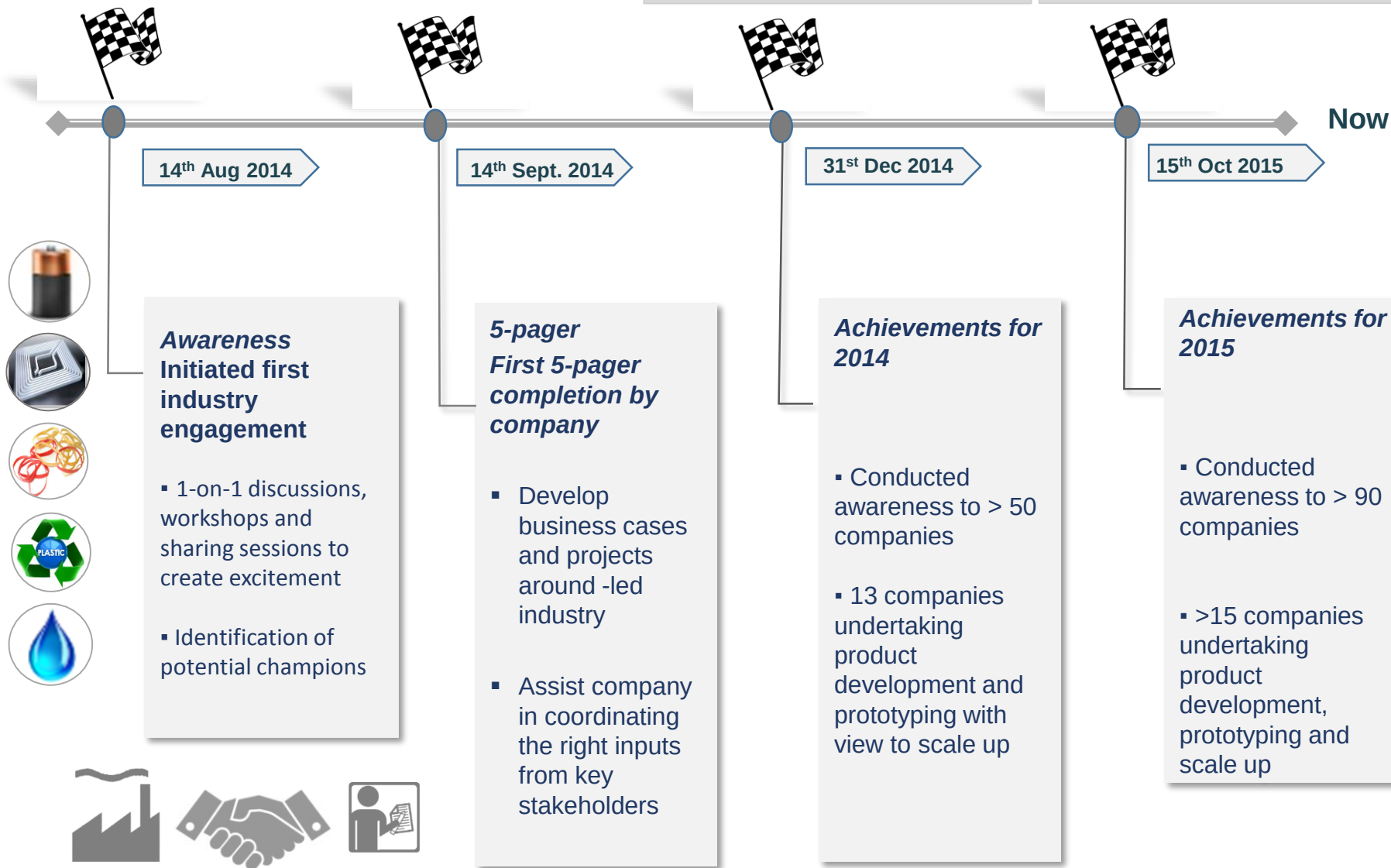
Nanofluids

# National Graphene Action Plan 2020 Milestones...



**2014**  
Awareness: > 50 co.  
Product Development: 9 Co.

**As of Nov 2015**  
Awareness: > 90 co.  
Product Development: 9 Co.



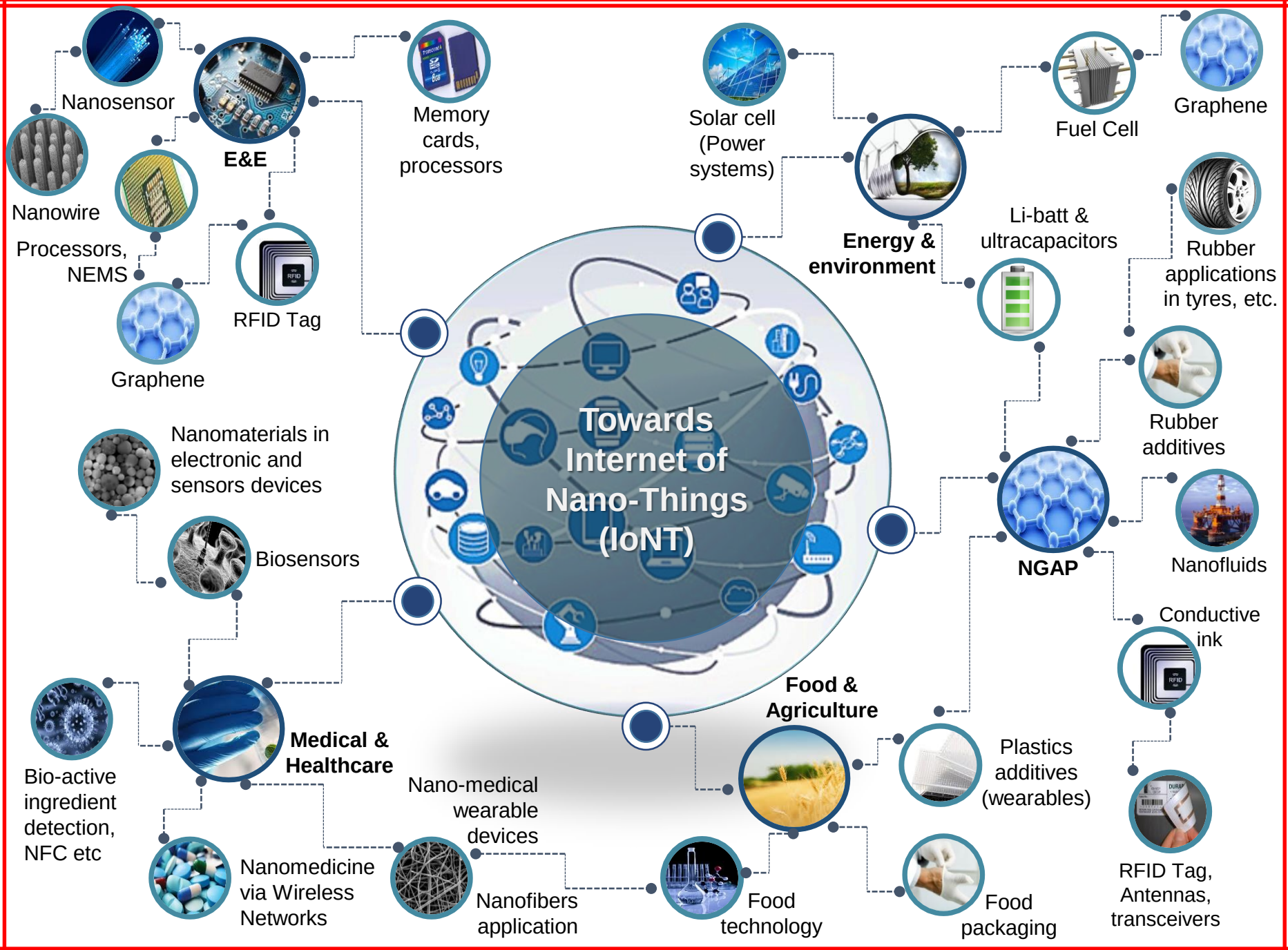
# Projects Identified Under The NGAP2020 Program

|   | Year | Business Nature                                                      | Application                                                                                                           | Project                                                                                                                                 |
|---|------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 2014 | Polypropylene masterbatch manufacturer                               | Plastics additives                 | Development of graphene-based polypropylene with high horizontal thermal conductivity for solar applications                            |
| 2 |      | Largest rubber glove manufacturer                                    | Rubber additives                   | Development of thin and high conductivity gloves for ESD application                                                                    |
| 3 |      | Solution provider for digital identity (smart card, etc.)            | Conductive inks                    | Development for RFID applications using conductive inks                                                                                 |
| 4 |      | Developer, manufacturer & marketer of rubber related compound        | Rubber additives                   | The prototyping of graphene-based technical compounds, rethreading and tyre compounds with high performance for automotive applications |
| 5 |      | Advanced polymer solution provider for automotive, electronics, etc. | Conductive inks                    | The prototyping of low cost graphene conductive inks for 3D printing and solar photovoltaics industry                                   |
| 6 |      | Leader in rubber recycling technology                                | Rubber additives                   | The prototyping of graphene-enabled masterbatch for rubber industry with superior rolling, abrasion and thermal resistance              |
| 7 |      | Ultracapacitor developer and manufacturer                            | Li-ion battery/ ultracapacitors   | The scale-up production of ultracapacitors and R&D on Lithium-capacitors (Li-cap) for automotive and solar industry                     |
| 8 |      | Graphene-based quantum cell for automotive and solar application     | Li-ion Battery/ ultracapacitors  | The scale-up of graphene-based automotive ultracapacitors with lithium-ion batteries                                                    |
| 9 |      | Renowned male contraceptive device manufacturer                      | Rubber additives                 | To produce the world's thinnest male contraceptive device with enhanced comfort and outstanding safety aspect via graphene technology   |

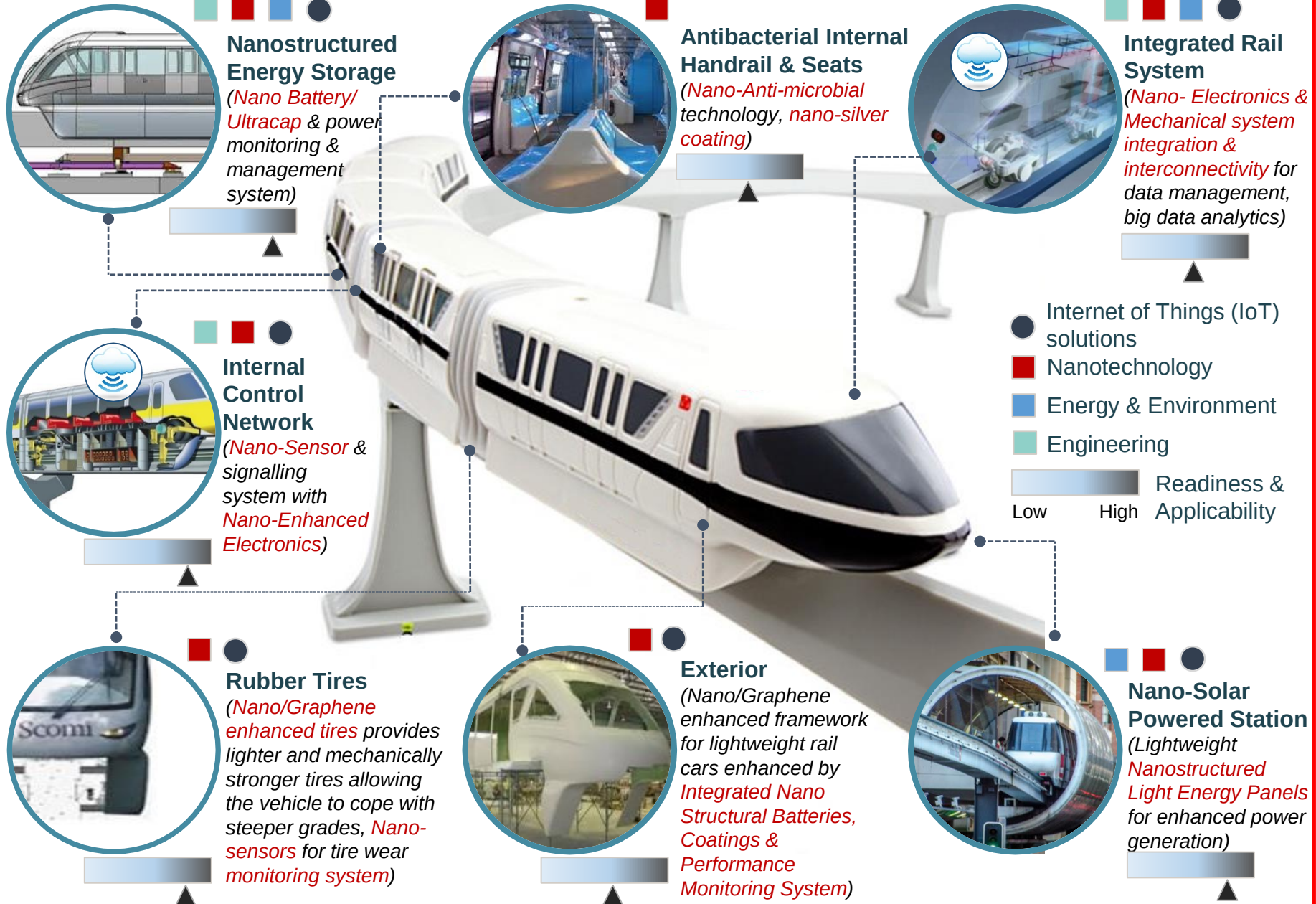


# Projects Identified Under The NGAP2020 Program

|    | Year | Business Nature                                 | Application                                                                                                           | Project                                                                                   |
|----|------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 10 | 2015 | Automobile manufacturer                         | Plastics additives                 | Development of graphene-based reinforced polymers nanocomposites for automotive component |
| 11 |      | Electric Vehicle manufacturer                   | Li-ion battery/ ultracapacitors    | Development of graphene-based plate cover of battery module                               |
| 12 |      | Rubber hose manufacturer                        | Rubber additives                   | Developing graphene-based hose for improved mechanical properties and flame retardancy    |
| 13 |      | Plastic manufacturing solutions                 | Plastics additives                 | Developing graphene-based component for thermoplastic engineering polymer                 |
| 14 |      | Manufacturer of Wing-In-Ground aircraft         | Plastics additives                 | Developing graphene-based composite for Wing-In-Ground aircraft's body application        |
| 15 |      | Nitrile glove manufacturer                      | Plastics additives                 | Developing graphene-based nitrile glove for improved mechanical properties                |
| 16 |      | Specialty lubricants manufacturer               | Nanofluids                       | Developing graphene-based lubricants for improved thermal conductivity properties         |
| 17 |      | Self-adhesive tapes manufacturer                | Conductive inks                  | Developing graphene-based conductive ink on tape for electronic application               |
| 18 |      | Advanced materials developer for energy storage | Li-ion Battery/ ultracapacitors  | Developing silicone graphene anodes for the next generation of lithium-ion batteries      |



# NanoMalaysia IoNT Program 1 : Smart Transportation





# The End Game...

## 1 Bridging the gap:

- Industry and Research Academia
- Government research agencies and universities in Malaysia with existing prototyping facilities



Research Agencies



Government



Universities

## 2 Industry

- Encourage industry to move up the value chain by introducing high technology products
- Increase market share (local and global)
- New market penetration
- Retain market competitiveness



## 3 National

By 2020, the National Graphene Action Plan has the potential to add more than RM 20 Bn in GNI impact and help create 9,000 new Malaysian jobs

- 20-30% of these jobs expected to be high value - e.g. R&D, business development etc.



# THANK YOU

**Contact : Dr. Daniel Bien**  
**[daniel.bien@nanomalaysia.com.my](mailto:daniel.bien@nanomalaysia.com.my)**

**NANOMALAYSIA BERHAD** (955265-P)

(A CLG under the Ministry of Science, Technology and Innovation)

A-2-2, Level 2, 157 Hampshire Place Office,

No. 1, Jalan Mayang Sari, 50450 Kuala Lumpur.

Tel : +603 2166 8849

Fax : +603 2181 8849

Email : [info@nanomalaysia.com.my](mailto:info@nanomalaysia.com.my)

Website : [www.nanomalaysia.com.my](http://www.nanomalaysia.com.my)

Facebook : [www.facebook.com/NanoMalaysiaBerhad](https://www.facebook.com/NanoMalaysiaBerhad)

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Instagram : [@nanomalaysia](https://www.instagram.com/nanomalaysia) #nanomalaysia #nanoforall



# Alignment with NKEAs of Economic Transformation Programme (ETP)

| NKEA                                                                                                                     | Applications                                                                                                         | EPP                                                                     | EPP Objectives                                                                                                                                    | Graphene's role                                                                  |
|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Electronics &amp; Electrical</b><br> | <i>All</i><br>                      | EPP 20: Enabling Industries through Nanotechnology                      | Focuses on the application of nanotechnology, which has the potential to impact all 12 NKEAs in varying degrees                                   | Graphene is one of the most promising nanotechnology materials                   |
|                                                                                                                          | Li-ion battery anode / ultracap<br> | EPP 18: Enabling Electric Vehicle Component Manufacturing               | Fast-track the implementation of the Electric Vehicles (EV) policy; includes establishing a Lithium-Ion (Li-Ion) battery manufacturer in Malaysia | Graphene enables the next generation of fast-charging, high capacity batteries   |
|                                                                                                                          | Conduc-<br>tive<br>Inks<br>         | EPP 7: Increasing Solar Module Producers                                | Secure multinational module producers and facilitate the establishment of another five joint ventures                                             | Graphene inks helps companies reduce production costs                            |
| <b>Oil, Gas, and Energy</b><br>        | Plas-<br>tics<br>                   | EPP 13: Increase Petrochemical Outputs                                  | Help PETRONAS expand its petrochemical business through RAPID project in Johor and SAMUR project in Sabah                                         | Graphene can enable higher performing or cheaper plastics                        |
|                                                                                                                          | Nano-<br>fluids<br>               | EPP 1: Rejuvenating Existing Fields through Enhanced Oil Recovery (EOR) | Encourage the use of Enhanced Oil Recovery (EOR) –to improve oil recovery from industry norms                                                     | Graphene-enabled nanofluids increase the efficiency of oil extraction operations |
| <b>Palm Oil &amp; Rubber</b><br>      | Rubber<br>                        | EPP 9.3: Increase World Market Share of Latex Gloves to 65% by 2020     | Boost the natural rubber glove industry's global market share from 62% in 2011 to 65% by 2020                                                     | Graphene can be used to make high-quality gloves that are thinner and tougher    |